HAZARD AND OPERABILITY (HAZOP) STUDY REPORT

Hexane Storage Tank T-301

Hexane Storage Unit – Equipment-Based Study (Node 1)

SYNTHETIC DOCUMENT – FOR EXPERIMENTAL USE ONLY

This report was compiled from a semi-synthetic data set. It is *not* the record of an actual HAZOP workshop and must not be used for design, operation, safety assessment, permitting or any other real-world decision.

Document number	SYN-HAZOP-T301-001
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Data source	T301_HAZOP_CCPS2001_style.csv (94 records)

Document control

Revision history

Rev.	Date	Description	Prepared	Checked	Approved
A	4 Nov 2026	Initial synthetic issue compiled from the 94-record data set	[to be completed]	[to be completed]	[to be completed]

Study team

No HAZOP workshop was held for this document. The roles below are those named in Single et al. (2020) for a typical HAZOP team and are provided as a template only.

Role	Name	Attended
Study leader	[not applicable – synthetic]	–
Process engineer	[not applicable – synthetic]	–
Process control engineer	[not applicable – synthetic]	–
Operator	[not applicable – synthetic]	–
Safety engineer	[not applicable – synthetic]	–

Distribution

Experimental use by the compiler only.

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1 Experimental redaction notice

This is Version C, a controlled incomplete variant of Version A. Consequence fields and linked recommendation text have been deliberately withheld for 12 selected synthetic scenarios. The omissions are experimental interventions, not missing values in an authentic plant record. Do not use this document for operational or safety decisions. The complete values are stored separately in a ground-truth file and must not be supplied to the extraction/recovery model during the test.

2 Executive summary

This report presents a guideword-based HAZOP study of hexane storage tank T-301 and its associated inlet, outlet, vent, drain and transfer equipment, treated as a single node. The worksheet comprises 94 scenarios across eight deviations. 40 scenarios are reconstructed from the scenario chains reported by Single et al. (2020) for the hexane tank case of CCPS (2001); the remaining 54 are synthetic continuations generated to extend coverage of each deviation. The two groups are kept distinguishable throughout.

Item	Value
Nodes studied	1
Deviations examined (guideword–parameter pairs)	8
Scenarios recorded	94
reconstructed from the reference case	40
synthetic continuation	54
Scenarios with a fire, explosion or ignition outcome[a]	36
Scenarios with a risk ranking assigned	0
Recommendations recorded (all from synthetic records)	54

[a] Keyword count (fire, explosion, ignition) over the consequence field.

Principal points.

1. Loss of containment of a flammable liquid with a flash point of 251.15 K is the dominant hazard theme; 36 of 94 scenarios end in a fire, explosion or ignition outcome.
2. **No scenario has been risk ranked.** Every record carries the status *not assessed*, so the findings cannot yet be prioritised and recommendation priorities are shown as *TBD*.
3. The reconstructed scenarios list safeguard *concepts* from the reference ontology, whose installation status is unverified. The synthetic scenarios record no existing safeguards; the 54 recommendations therefore mostly ask for protective functions to be reviewed, verified or provided.
4. The conservation vent and dike shown on the reference P&ID are outside the modelled node in the source study and should be confirmed explicitly in any follow-on, node-based study.

3 Introduction

3.1 Purpose and objectives

The purpose of this document is to present, in the form of a formal HAZOP report, the deviations, causes, consequences, safeguards and recommendations recorded for tank T-301. The objectives are to (i) document each deviation examined and the scenarios associated with it, (ii) show which scenarios

originate in the reference case and which are synthetic, and (iii) consolidate the recommendations into a register suitable for follow-up.

3.2 Scope

The study covers a single node: hexane storage tank T-301 together with its inlet and outlet valves, level control loop, drain line and valve, vent arrangement, and the transfer (recirculation) pump 4-41. Upstream tank-truck unloading and the downstream consumer are not studied in detail. Propagation of deviations between nodes is outside the scope.

3.3 Nature of this document

This is a synthetic report prepared for an experiment. Three consequences follow and are repeated where they matter:

- **Provenance.** Rows marked `P` are reconstructions of scenario chains reported in the literature, not verbatim rows of a worksheet. Rows marked `S` are newly generated and are not attributed to either source publication.
- **Safeguards.** For synthetic scenarios, no existing safeguard is established and none may be assumed. For reconstructed scenarios, safeguards are ontology concepts reported in the source and are marked with a dagger (†); their installation status requires verification.
- **Risk.** No likelihood or severity assessment has been made.

4 Process description

4.1 Overview

Tank T-301 stores liquid hexane, which is typically used as a solvent or reaction medium. The source describes the tank as pressurised owing to the vapour pressure of hexane, located outdoors, and normally about half full. Level is controlled by a control loop with a level transmitter (LT) and level indicator controller (LIC); the source text also refers to a “pressure indicator controller” in this context, and this report follows the P&ID designation. Pump 4-41 supplies a downstream process that is not further specified. The tank is considered in isolation, so the upstream process is likewise unspecified.

4.2 Principal equipment

Item	Description (per source P&ID and model)
T-301	Hexane storage tank; storage of liquid hexane is the intended function
Inlet valve / unloading line	Receipt from the tank truck through the hexane unloading pump
Outlet valve	Outlet to the transfer pump
Pump 4-41	Hexane pump supplying the downstream process; modelled as the recirculation/transfer pump
LT / LIC	Level transmitter and level indicator controller forming the level loop
Drain line and valve	Low-point drain of the tank
Conservation vent to dike	Shown on the P&ID; deliberately not modelled as part of the tank
Dike	Secondary containment; deliberately not modelled as part of the tank

4.3 Substance

Hexane is handled as a flammable liquid with a flash point of 251.15 K (−22.0 °C), as modelled in the source ontology.

4.4 Study assumptions

The following circumstances are taken from the source study:

- ignition sources are assumed to be present unless they can be excluded;
- ambient air is present because the tank is located outdoors, with sufficient oxygen available;
- an operator can interact with the process;
- impurities, including water, may be introduced, for example during filling.

4.5 Safeguards reported in the reference worksheet

For context only, the original CCPS (2001) worksheet, as reproduced by Single et al. (2020), reports the following safeguards and recommendations for this tank: a high-level alarm audible in the control room; a hexane unloading procedure with a checklist; standard steam-out procedures; operation and maintenance response including isolation; the ability to isolate the tank manually; periodic non-destructive inspection per API recommended practice and ASME code; a relief valve discharging to the tank's dike; a dike sized for 1.5 times the tank capacity; and emergency response procedures. These are *not* carried into the worksheet below as installed safeguards.

5 Methodology

5.1 HAZOP procedure

The study follows the standard scenario-based HAZOP sequence summarised by Single et al. (2020): (0) collect the process-safety information; (1) divide the plant into nodes using the P&ID or PFD; (2) establish the design intent of each node; (3) apply deviations to the node; (4) discuss potential causes and consequences; and (5) identify safeguards and recommendations. Where required, likelihood and severity are then assessed. The framework is consistent with IEC 61882, although formal conformance is not claimed.

5.2 Node and design intent

Node 1: T-301 hexane storage tank and associated inlet/outlet, vent, drain and transfer equipment.
Design intent: receive and store liquid hexane while maintaining containment and allowing controlled transfer.

5.3 Deviations and nomenclature

Reconstructed records use compound deviation names (for example *HighLevel*), while synthetic records use guideword–parameter pairs. This report harmonises both into the eight pairs below. The reference ontology's *elsewhere flow* deviation is presented as loss of containment.

Deviation	Guideword	Parameter	P	S	Total
High level	MORE	Level	6	5	11
Low level	LESS	Level	6	4	10
High pressure	MORE	Pressure	6	8	14
Low pressure	LESS	Pressure	6	8	14
High temperature	MORE	Temperature	4	6	10
Low temperature	LESS	Temperature	2	6	8
Other than composition	OTHER THAN	Composition	4	9	13
Elsewhere flow / loss of containment	OTHERWISE/ELSEWHERE	Flow	6	8	14
Total			**40**	**54**	**94**

P = reconstructed from the reference case; S = synthetic continuation.

5.4 Record conventions

In the cause column of reconstructed records, the *cause* is the immediate cause and the *super cause* is the higher-level cause behind it. In the consequence column, the *effect* is the immediate physical result of the deviation and the *consequence* is the chain that follows, shown with arrows. Rows of synthetic scenarios are shaded. The risk-ranking column reads *not assessed* for all records.

6 HAZOP worksheets

Each table lists the scenarios for one deviation of Node 1. P marks a reconstruction of a scenario chain reported in Single et al. (2020); S marks a synthetic scenario (shaded). Safeguards marked † are ontology concepts reported in the source; their installation status is unverified.

6.1 High level

Guideword: MORE **Parameter:** Level **Scenarios:** 11 (6 reconstructed, 5 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-001	P	*Cause:* Excessive inflow	*Effect:* Overfilling *Consequence:* Loss of primary containment → Fire	Collecting basin, overfill protection†	None mapped from source; domain-expert review required	Not assessed
T301-002	P	*Cause:* Excessive inflow	*Effect:* Overfilling *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, overfill protection†	None mapped from source; domain-expert review required	Not assessed
T301-003	P	*Cause:* Closed outlet valve *Super cause(s):* Operational error, valve failure	*Effect:* Overfilling *Consequence:* Loss of primary containment → Fire	Collecting basin, overfill protection†	None mapped from source; domain-expert review required	Not assessed
T301-004	P	*Cause:* Closed outlet valve *Super cause(s):* Operational error, valve failure	*Effect:* Overfilling *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, overfill protection†	None mapped from source; domain-expert review required	Not assessed
T301-005	P	*Cause:* Malfunction flow controller	*Effect:* Overfilling *Consequence:* Loss of primary containment → Fire	Collecting basin, overfill protection†	None mapped from source; domain-expert review required	Not assessed
T301-006	P	*Cause:* Malfunction flow controller	*Effect:* Overfilling *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, overfill protection†	None mapped from source; domain-expert review required	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-041	S	Unloading continues after target inventory; incorrect inventory estimate; level controller fails to reduce inflow.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-042	S	Inlet valve fails open; transfer pump continues running; shutdown signal fails.	Overfill and loss of containment; potential pool fire or vapor-cloud ignition.	None established (not assumed present)	Review fail-safe inlet isolation and independent high-high level trip.	Not assessed
T301-043	S	Level instrument under-reads; calibration drift; impulse connection obstructed.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-044	S	Outlet line blocked or outlet valve closed while inflow continues.	Inventory accumulates; tank may overfill; released hexane can create a flammable pool/vapor.	None established (not assumed present)	Interlock or procedural control for simultaneous inflow with unavailable outlet; add high-level alarm.	Not assessed
T301-045	S	Transfer routed to T-301 when another tank was intended.	Unexpected inventory accumulation and potential overfill.	None established (not assumed present)	Use positive tank/line identification and pre-transfer lineup verification.	Not assessed

6.2 Low level

Guideword: LESS **Parameter:** Level **Scenarios:** 10 (6 reconstructed, 4 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-007	P	*Cause:* Blocked inflow line *Super cause(s):* Deposition of impurities	*Effect:* Pump running dry *Consequence:* Pump breakdown → Production downtime	Dry run protection, level controller low alarm†	None mapped from source; domain-expert review required	Not assessed
T301-008	P	*Cause:* Closed inlet valve *Super cause(s):* Human error, operational error, valve failure	*Effect:* Pump running dry *Consequence:* Pump breakdown → Production downtime	Dry run protection, level controller low alarm†	None mapped from source; domain-expert review required	Not assessed
T301-009	P	*Cause:* Closed inlet valve *Super cause(s):* Human error, operational error, valve failure	*Effect:* Emptying of container *Consequence:* Production downtime	Level controller low alarm†	None mapped from source; domain-expert review required	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-010	P	*Cause:* Malfunction flow controller	*Effect:* Emptying of container *Consequence:* Production downtime	Level controller low alarm†	None mapped from source; domain-expert review required	Not assessed
T301-011	P	*Cause:* Loss of inflow *Super cause(s):* Operational error	*Effect:* Pump running dry *Consequence:* Pump breakdown → Production downtime	Dry run protection, level controller low alarm†	None mapped from source; domain-expert review required	Not assessed
T301-012	P	*Cause:* Loss of inflow *Super cause(s):* Operational error	*Effect:* Emptying of container *Consequence:* Production downtime	Level controller low alarm†	None mapped from source; domain-expert review required	Not assessed
T301-046	S	Transfer pump continues drawing liquid; inflow is lower than outflow.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-047	S	Upstream supply interruption; inlet valve fails closed; inlet strainer blocks.	Tank empties during continued withdrawal; transfer interruption and possible pump damage.	None established (not assumed present)	Add low-level alarm and verify pump protection for loss of suction.	Not assessed
T301-048	S	Level instrument over-reads and controller reduces or stops inflow.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-049	S	Tank, drain, flange, or connected line leaks while withdrawal continues.	Loss of hexane containment, falling inventory, flammable vapor and potential ignition.	None established (not assumed present)	Inspect tank and joints; verify leak detection, isolation, and secondary containment.	Not assessed

6.3 High pressure

Guideword: MORE **Parameter:** Pressure **Scenarios:** 14 (6 reconstructed, 8 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-023	P	*Cause:* Blocked outflow line *Super cause(s):* Deposition of impurities	*Effect:* Crack *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure relief valve†	None mapped from source; domain-expert review required	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-024	P	*Cause:* Blocked outflow line *Super cause(s):* Deposition of impurities	*Effect:* Crack *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, pressure relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-025	P	*Cause:* Blocked outflow line *Super cause(s):* Deposition of impurities	*Effect:* Rupture *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure relief valve, pressure vacuum relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-026	P	*Cause:* Blocked outflow line *Super cause(s):* Deposition of impurities	*Effect:* Rupture *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, pressure relief valve, pressure vacuum relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-027	P	*Cause:* Thermal expansion *Super cause(s):* Abnormal hot intake, external fire, solar radiation	*Effect:* Rupture *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure relief valve, pressure vacuum relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-028	P	*Cause:* Thermal expansion *Super cause(s):* Abnormal hot intake, external fire, solar radiation	*Effect:* Rupture *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, pressure relief valve, pressure vacuum relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-067	S	Outlet or vent path obstructed; vapor generation continues.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-068	S	Liquid warms while the tank or connected section is isolated and has insufficient expansion space.	Pressure rise in trapped liquid or vapor space; leakage or mechanical failure possible.	None established (not assumed present)	Review blocked-in liquid thermal expansion and ensure appropriate protection.	Not assessed
T301-069	S	External fire heats tank contents and increases vapor generation.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-070	S	Incorrect tie-in or higher-pressure source connected to tank.	Tank overpressure if not designed for source pressure; leakage or rupture.	None established (not assumed present)	Verify connection compatibility and prevent unintended pressurization.	Not assessed
T301-071	S	Instrument reads low or control action fails to respond.	High pressure may remain undetected until release or equipment damage.	None established (not assumed present)	Review pressure monitoring and independent protective functions where justified.	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-072	S	Vent path partly obstructed by debris, icing, or maintenance configuration.	Pressure increase during filling or warming; potential loss of containment.	None established (not assumed present)	Inspect vent paths and verify capacity for relevant operating scenarios.	Not assessed
T301-073	S	Tank is filled beyond intended operating inventory.	Liquid expansion or continued inflow can cause release; tank pressure effects depend on configuration.	None established (not assumed present)	Review maximum fill basis and independent overfill prevention.	Not assessed
T301-074	S	Unapproved connection or component changes tank pressure behavior.	Overpressure or leakage under an unintended pressure source.	None established (not assumed present)	Apply management of change and verify pressure ratings after modifications.	Not assessed

6.4 Low pressure

Guideword: LESS **Parameter:** Pressure **Scenarios:** 14 (6 reconstructed, 8 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-029	P	*Cause:* Ambient temperature change	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure vacuum relief valve, vacuum proof design†	None mapped from source; domain-expert review required	Not assessed
T301-030	P	*Cause:* Ambient temperature change	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, pressure vacuum relief valve, vacuum proof design, define Ex protection zone†	None mapped from source; domain-expert review required	Not assessed
T301-031	P	*Cause:* Obstructed vent path *Super cause(s):* Faulty installation, human error	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure vacuum relief valve, vacuum proof design†	None mapped from source; domain-expert review required	Not assessed
T301-032	P	*Cause:* Obstructed vent path *Super cause(s):* Faulty installation, human error	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, pressure vacuum relief valve, vacuum proof design†	None mapped from source; domain-expert review required	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-033	P	*Cause:* Excessive fluid withdrawal *Super cause(s):* Malfunction transfer pump	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Fire	Collecting basin, pressure vacuum relief valve, vacuum proof design†	None mapped from source; domain-expert review required	Not assessed
T301-034	P	*Cause:* Excessive fluid withdrawal *Super cause(s):* Malfunction transfer pump	*Effect:* Collapse of enclosure *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, vacuum proof design, define Ex protection zone, pressure vacuum relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-075	S	Tank cools while vapor condenses and make-up gas/venting is insufficient.	Vacuum develops; tank may buckle or collapse if not vacuum-rated.	None established (not assumed present)	Verify vacuum protection and cooldown procedure against tank design.	Not assessed
T301-076	S	Vent cannot admit sufficient make-up gas as liquid is withdrawn.	Vacuum develops; tank deformation or collapse may cause loss of containment.	None established (not assumed present)	Inspect vent path and verify vacuum relief capacity for withdrawal.	Not assessed
T301-077	S	Rapid cooling condenses vapor in a tank with limited make-up gas.	Vacuum loading and potential tank collapse.	None established (not assumed present)	Assess condensation/cooldown case and vacuum protection.	Not assessed
T301-078	S	Transfer pump removes liquid faster than make-up gas enters.	Vacuum condition; possible structural damage if tank is not designed for it.	None established (not assumed present)	Review maximum withdrawal rate and venting capacity.	Not assessed
T301-079	S	Tank is isolated during cooldown after hot cleaning or steam-out.	Vacuum formation and possible enclosure collapse.	None established (not assumed present)	Use approved cleaning/cooldown procedure and verify vacuum protection.	Not assessed
T301-080	S	Blanketing/make-up gas source is unavailable where such a system is used.	Vacuum may develop during withdrawal or cooling; air ingress may alter vapor-space conditions.	None established (not assumed present)	If blanketing is part of the design, alarm on supply failure and verify independent vacuum protection.	Not assessed
T301-081	S	Pressure instrument is biased or blocked.	Operator may fail to recognize vacuum condition.	None established (not assumed present)	Inspect and test pressure instruments; use independent verification for critical indication.	Not assessed
T301-082	S	Tank and vent/make-up paths isolated while contents cool.	Vacuum formation and possible structural damage.	None established (not assumed present)	Review isolation line-up and provide a safe cooldown path.	Not assessed

6.5 High temperature

Guideword: MORE **Parameter:** Temperature **Scenarios:** 10 (4 reconstructed, 6 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-035	P	*Cause:* Heat input by recirculation pump *Super cause(s):* Mechanical damage	*Effect:* Abnormal evaporation *Consequence:* Exceeding allowable pressure	Pressure relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-036	P	*Cause:* Heat input by recirculation pump *Super cause(s):* Mechanical damage	*Effect:* Exceeding flash point *Consequence:* none identified	Ignition source assessment†	None mapped from source; domain-expert review required	Not assessed
T301-037	P	*Cause:* Abnormal heat input *Super cause(s):* Abnormal hot intake, external fire, solar radiation	*Effect:* Abnormal evaporation *Consequence:* Exceeding allowable pressure	Pressure relief valve†	None mapped from source; domain-expert review required	Not assessed
T301-038	P	*Cause:* Abnormal heat input *Super cause(s):* Abnormal hot intake, external fire, solar radiation	*Effect:* Exceeding flash point *Consequence:* none identified	Ignition source assessment†	None mapped from source; domain-expert review required	Not assessed
T301-083	S	Prolonged solar exposure raises stored liquid temperature.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-084	S	Tank exposed to nearby fire or radiant heat.	Rapid heating and vapor generation; pressure rise; possible loss of containment.	None established (not assumed present)	Review fire exposure, emergency response, and applicable tank protection.	Not assessed
T301-085	S	Unexpectedly hot liquid transferred into tank.	Temperature excursion, increased vapor generation, possible pressure rise.	None established (not assumed present)	Verify incoming material temperature where relevant and define transfer acceptance conditions.	Not assessed
T301-086	S	Mechanical fault or prolonged recirculation adds heat to stored liquid.	Increased evaporation and vapor pressure; potential pressure-control challenge.	None established (not assumed present)	Monitor pump condition and evaluate heat input during abnormal operation.	Not assessed
T301-087	S	Transfer or recirculation occurs under unintended thermal conditions.	Unexpected vapor generation and pressure increase.	None established (not assumed present)	Review operating sequence and temperature monitoring needs.	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-088	S	Temperature instrument under-reads actual liquid temperature.	Abnormal heating may go unnoticed; delayed response to rising vapor generation.	None established (not assumed present)	Verify temperature indication and calibration; use independent checks where warranted.	Not assessed

6.6 Low temperature

Guideword: LESS **Parameter:** Temperature **Scenarios:** 8 (2 reconstructed, 6 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-039	P	*Cause:* Ambient temperature change	*Effect:* Brittle fracture *Consequence:* Loss of primary containment → Fire	Collecting basin†	None mapped from source; domain-expert review required	Not assessed
T301-040	P	*Cause:* Ambient temperature change	*Effect:* Brittle fracture *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone†	None mapped from source; domain-expert review required	Not assessed
T301-089	S	Cold weather cools the tank or connected drain line.	Material/metal may be exposed to low-temperature conditions; water in drains may freeze; brittle fracture is possible if material suitability is inadequate.	None established (not assumed present)	Check minimum design metal temperature/material suitability and protect vulnerable drains.	Not assessed
T301-090	S	Colder-than-intended material enters tank.	Thermal excursion; potential material embrittlement concerns depending on design and material.	None established (not assumed present)	Verify temperature compatibility and review material design limits.	Not assessed
T301-091	S	Cold utility contacts a process boundary due to exchanger or line failure.	Localized cooling; possible brittle fracture if below material limits.	None established (not assumed present)	Inspect heat-transfer boundaries and assess low-temperature material suitability.	Not assessed
T301-092	S	Temperature instrument reads higher than actual temperature.	Cold condition remains undetected; freezing or embrittlement concerns may be missed.	None established (not assumed present)	Calibrate temperature instrumentation and assess independent monitoring.	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-093	S	Water remains in drain or low point and ambient temperature falls.	Ice may obstruct or damage drain line; leakage may follow if the line fractures.	None established (not assumed present)	Verify drainage and inspect/maintain low-point drains for the expected environment.	Not assessed
T301-094	S	Insulation or weather protection is damaged or absent.	Cold spots; possible embrittlement or water freezing depending on materials and contents.	None established (not assumed present)	Inspect weather protection and verify material suitability for ambient conditions.	Not assessed

6.7 Other than composition

Guideword: OTHER THAN **Parameter:** Composition **Scenarios:** 13 (4 reconstructed, 9 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-013	P	*Cause:* Contaminants	*Effect:* Poor product quality *Consequence:* No safety consequence	Periodical sample taking†	None mapped from source; domain-expert review required	Not assessed
T301-014	P	*Cause:* Confusion of substances *Super cause(s):* Human error	No effect or consequence identified in source	Periodical sample taking†	None mapped from source; domain-expert review required	Not assessed
T301-015	P	*Cause:* Contamination by water *Super cause(s):* Condensation air humidity, introduction of water	*Effect:* Drainline fracture *Consequence:* Loss of primary containment → Fire	Water detection system, collecting basin, periodical sample taking†	None mapped from source; domain-expert review required	Not assessed
T301-016	P	*Cause:* Contamination by water *Super cause(s):* Condensation air humidity, introduction of water	*Effect:* Drainline fracture *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Water detection system, collecting basin, define Ex protection zone†	None mapped from source; domain-expert review required	Not assessed
T301-050	S	Water enters through an open connection, condensation, or incomplete draining after cleaning.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-051	S	Wrong substance transferred; contaminated delivery; residue remains in tank.	Off-spec inventory; unexpected material properties; possible incompatibility depending on contaminant.	None established (not assumed present)	Require delivery identification and representative incoming-material sampling.	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-052	S	Transfer connection or tank selection error.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-053	S	Tank or line returned to service without adequate draining/clearance.	Off-spec hexane; potential unexpected interaction with residues.	None established (not assumed present)	Introduce documented line/tank clearance and return-to-service checks.	Not assessed
T301-054	S	Water remains after washing or enters through an open drain connection.	Potential freezing at low temperature; drain-line blockage or fracture; possible loss of containment.	None established (not assumed present)	Verify drainage and inspect drain line condition before cold-weather operation.	Not assessed
T301-055	S	Supplier quality variation or inadequate receipt checks.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-056	S	Sample mislabeled, contaminated, or taken from an unrepresentative point.	Wrong operational decision; contaminated batch may be accepted.	None established (not assumed present)	Use sample labeling, representative sampling, and repeat testing for anomalous results.	Not assessed
T301-057	S	Tank opened or vent/drain connection left improperly configured.	Potential change in vapor-space conditions; moisture contamination; flammable atmosphere may still exist depending on conditions.	None established (not assumed present)	Review opening/closing procedure and verify connection status after maintenance.	Not assessed
T301-058	S	Water ingress, contamination, or unsuitable storage conditions.	Product quality deterioration; downstream operating issues.	None established (not assumed present)	Establish periodic inspection and sampling frequency based on material specification.	Not assessed

6.8 Elsewhere flow / loss of containment

Guideword: OTHERWISE/ELSEWHERE **Parameter:** Flow **Scenarios:** 14 (6 reconstructed, 8 synthetic)

ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-017	P	*Cause:* Mechanical seal failure of pump *Super cause(s):* Improper installation, operational error	*Effect:* Pool formation *Consequence:* Pump breakdown → Production downtime	Collecting basin†	None mapped from source; domain-expert review required	Not assessed
T301-018	P	*Cause:* Mechanical seal failure of pump *Super cause(s):* Improper installation, operational error	*Effect:* Pool formation *Consequence:* Loss of primary containment → Fire	Collecting basin†	None mapped from source; domain-expert review required	Not assessed
T301-019	P	*Cause:* Mechanical seal failure of pump *Super cause(s):* Improper installation, operational error	*Effect:* Pool formation *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, explosion proof motor†	None mapped from source; domain-expert review required	Not assessed
T301-020	P	*Cause:* External leakage *Super cause(s):* Corrosion, erosion, leaky connections, material defect, mechanical damage	*Effect:* Pool formation *Consequence:* Pump breakdown → Production downtime	Collecting basin†	None mapped from source; domain-expert review required	Not assessed
T301-021	P	*Cause:* External leakage *Super cause(s):* Corrosion, erosion, leaky connections, material defect, mechanical damage	*Effect:* Pool formation *Consequence:* Loss of primary containment → Fire	Collecting basin†	None mapped from source; domain-expert review required	Not assessed
T301-022	P	*Cause:* External leakage *Super cause(s):* Corrosion, erosion, leaky connections, material defect, mechanical damage	*Effect:* Pool formation *Consequence:* Loss of primary containment → Formation of Ex atmosphere → Explosion	Collecting basin, define Ex protection zone, explosion proof motor†	None mapped from source; domain-expert review required	Not assessed
T301-059	S	Seal wear, incorrect installation, shaft misalignment, or vibration.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed
T301-060	S	Gasket deterioration, bolt loosening, corrosion, or poor assembly.	Hexane release near connection; pool fire or flash fire possible after ignition.	None established (not assumed present)	Inspect joints and verify gasket/material compatibility and assembly quality.	Not assessed
T301-061	S	Drain valve left open, damaged seat, or accidental operation.	[INTENTIONALLY OMITTED – CONSEQUENCE FIELD WITHHELD]	None established (not assumed present)	[WITHHELD – LINKED RECOMMENDATION NOT PROVIDED]	Not assessed

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ID	Src	Possible causes	Possible consequences	Existing safeguards	Recommendations	Risk ranking
T301-062	S	Corrosion, mechanical damage, or material defect.	Loss of primary containment; pool formation; possible fire or environmental release.	None established (not assumed present)	Implement condition inspection and repair criteria appropriate to tank design.	Not assessed
T301-063	S	Loose fitting, damaged tubing, or maintenance error.	Localized hexane release and potential flammable vapor accumulation.	None established (not assumed present)	Inspect instrument connections and consider isolation arrangements for small-bore fittings.	Not assessed
T301-064	S	Incorrect vent arrangement or liquid carryover during overfill.	Hexane release to atmosphere; flammable vapor cloud may form.	None established (not assumed present)	Review vent routing and overfill scenarios with qualified process-safety personnel.	Not assessed
T301-065	S	Incorrect seal or gasket installed; joint not properly tightened.	Hexane loss of containment during return to service.	None established (not assumed present)	Use maintenance quality checks and leak testing before return to service.	Not assessed
T301-066	S	Vehicle or handling equipment impacts tank or connected piping.	Rapid release of hexane; pool/vapor formation; fire/explosion potential.	None established (not assumed present)	Assess impact protection and emergency isolation needs.	Not assessed

7 Findings

1. **Ignition of released hexane dominates.** 36 of 94 scenarios record a fire, explosion or ignition outcome. Given the stated assumption that ignition sources are present, containment, vent routing and release control deserve first attention once risk is ranked.
2. **Risk ranking is outstanding.** All 94 scenarios are unranked, so no scenario can be called high or low risk, and the priority of all 54 recommendations is undetermined.
3. **Safeguard evidence is uneven.** Reconstructed scenarios cite ontology safeguard concepts (Appendix A) that have not been checked against an installed design. Synthetic scenarios cite none.
4. **Paired outcome rows.** Many reconstructed rows occur in pairs that share cause and effect and differ only in the consequence chain (fire versus explosion). A review team may wish to merge these.
5. **Scope boundaries.** The conservation vent and dike are outside the modelled node, and inter-node propagation is not examined.
6. **Recommendations are review-type.** The synthetic recommendations largely ask for protection to be reviewed, verified or provided, rather than naming an existing safeguard.

8 Recommendation register

The register consolidates the 54 recommendations recorded against synthetic scenarios, grouped by theme. None of the 40 reconstructed scenarios carries a mapped recommendation. Owners and priorities are not yet assigned.

* Priority cannot be set until scenarios have been risk ranked.

8.1 Level measurement, alarms and overfill / dry-run protection

Level detection, independent trip functions, and protection against overfill and loss of suction.

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-01	T301-041	High level	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-02	T301-042	High level	Review fail-safe inlet isolation and independent high-high level trip.	TBD	TBD	Open
R-03	T301-043	High level	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-04	T301-044	High level	Interlock or procedural control for simultaneous inflow with unavailable outlet; add high-level alarm.	TBD	TBD	Open
R-05	T301-046	Low level	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-06	T301-047	Low level	Add low-level alarm and verify pump protection for loss of suction.	TBD	TBD	Open
R-07	T301-048	Low level	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-08	T301-073	High pressure	Review maximum fill basis and independent overfill prevention.	TBD	TBD	Open

8.2 Transfer line-up, material identification and receipt sampling

Positive identification of tank, line and material, clearance before return to service, and acceptance sampling.

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-09	T301-045	High level	Use positive tank/line identification and pre-transfer lineup verification.	TBD	TBD	Open
R-10	T301-050	Other than composition	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-11	T301-051	Other than composition	Require delivery identification and representative incoming-material sampling.	TBD	TBD	Open

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-12	T301-052	Other than composition	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-13	T301-053	Other than composition	Introduce documented line/tank clearance and return-to-service checks.	TBD	TBD	Open
R-14	T301-055	Other than composition	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-15	T301-056	Other than composition	Use sample labeling, representative sampling, and repeat testing for anomalous results.	TBD	TBD	Open
R-16	T301-058	Other than composition	Establish periodic inspection and sampling frequency based on material specification.	TBD	TBD	Open

8.3 Vent, relief and vacuum protection

Vent and relief capacity, vacuum protection, overpressure sources, pressure instrumentation and management of change.

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-17	T301-064	Elsewhere flow / loss of containment	Review vent routing and overfill scenarios with qualified process-safety personnel.	TBD	TBD	Open
R-18	T301-067	High pressure	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-19	T301-068	High pressure	Review blocked-in liquid thermal expansion and ensure appropriate protection.	TBD	TBD	Open
R-20	T301-070	High pressure	Verify connection compatibility and prevent unintended pressurization.	TBD	TBD	Open
R-21	T301-071	High pressure	Review pressure monitoring and independent protective functions where justified.	TBD	TBD	Open
R-22	T301-072	High pressure	Inspect vent paths and verify capacity for relevant operating scenarios.	TBD	TBD	Open
R-23	T301-074	High pressure	Apply management of change and verify pressure ratings after modifications.	TBD	TBD	Open
R-24	T301-075	Low pressure	Verify vacuum protection and cooldown procedure against tank design.	TBD	TBD	Open
R-25	T301-076	Low pressure	Inspect vent path and verify vacuum relief capacity for withdrawal.	TBD	TBD	Open
R-26	T301-077	Low pressure	Assess condensation/cooldown case and vacuum protection.	TBD	TBD	Open
R-27	T301-078	Low pressure	Review maximum withdrawal rate and venting capacity.	TBD	TBD	Open
R-28	T301-079	Low pressure	Use approved cleaning/cooldown procedure and verify vacuum protection.	TBD	TBD	Open

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-29	T301-080	Low pressure	If blanketing is part of the design, alarm on supply failure and verify independent vacuum protection.	TBD	TBD	Open
R-30	T301-081	Low pressure	Inspect and test pressure instruments; use independent verification for critical indication.	TBD	TBD	Open
R-31	T301-082	Low pressure	Review isolation line-up and provide a safe cooldown path.	TBD	TBD	Open

8.4 Leak prevention, inspection, isolation and containment

Seal, joint, drain and instrument-connection integrity, maintenance quality, impact protection, leak detection and isolation.

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-32	T301-049	Low level	Inspect tank and joints; verify leak detection, isolation, and secondary containment.	TBD	TBD	Open
R-33	T301-057	Other than composition	Review opening/closing procedure and verify connection status after maintenance.	TBD	TBD	Open
R-34	T301-059	Elsewhere flow / loss of containment	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-35	T301-060	Elsewhere flow / loss of containment	Inspect joints and verify gasket/material compatibility and assembly quality.	TBD	TBD	Open
R-36	T301-061	Elsewhere flow / loss of containment	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-37	T301-062	Elsewhere flow / loss of containment	Implement condition inspection and repair criteria appropriate to tank design.	TBD	TBD	Open
R-38	T301-063	Elsewhere flow / loss of containment	Inspect instrument connections and consider isolation arrangements for small-bore fittings.	TBD	TBD	Open
R-39	T301-065	Elsewhere flow / loss of containment	Use maintenance quality checks and leak testing before return to service.	TBD	TBD	Open
R-40	T301-066	Elsewhere flow / loss of containment	Assess impact protection and emergency isolation needs.	TBD	TBD	Open

8.5 Thermal exposure and cold-weather protection

Heat input and fire exposure, temperature instrumentation, low-temperature material suitability and freeze protection of drains.

Ref.	Scenario	Deviation	Recommendation	Priority*	Owner	Status
R-41	T301-054	Other than composition	Verify drainage and inspect drain line condition before cold-weather operation.	TBD	TBD	Open
R-42	T301-069	High pressure	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-43	T301-083	High temperature	[WITHHELD FOR MISSING-LINK EVALUATION]	TBD	TBD	Open
R-44	T301-084	High temperature	Review fire exposure, emergency response, and applicable tank protection.	TBD	TBD	Open
R-45	T301-085	High temperature	Verify incoming material temperature where relevant and define transfer acceptance conditions.	TBD	TBD	Open
R-46	T301-086	High temperature	Monitor pump condition and evaluate heat input during abnormal operation.	TBD	TBD	Open
R-47	T301-087	High temperature	Review operating sequence and temperature monitoring needs.	TBD	TBD	Open
R-48	T301-088	High temperature	Verify temperature indication and calibration; use independent checks where warranted.	TBD	TBD	Open
R-49	T301-089	Low temperature	Check minimum design metal temperature/material suitability and protect vulnerable drains.	TBD	TBD	Open
R-50	T301-090	Low temperature	Verify temperature compatibility and review material design limits.	TBD	TBD	Open
R-51	T301-091	Low temperature	Inspect heat-transfer boundaries and assess low-temperature material suitability.	TBD	TBD	Open
R-52	T301-092	Low temperature	Calibrate temperature instrumentation and assess independent monitoring.	TBD	TBD	Open
R-53	T301-093	Low temperature	Verify drainage and inspect/maintain low-point drains for the expected environment.	TBD	TBD	Open
R-54	T301-094	Low temperature	Inspect weather protection and verify material suitability for ambient conditions.	TBD	TBD	Open

9 Limitations and open actions

1. Complete the risk ranking of all 94 scenarios using the organisation's risk matrix.
2. Verify each safeguard concept in Appendix A against the actual design.
3. Review the synthetic scenarios and recommendations with a qualified process-safety team.
4. Extend the study to a node-based analysis including the vent, dike, and upstream and downstream units, and examine propagation of deviations between nodes.
5. Resolve the descriptive difference between the level-loop wording in the source text and its P&ID.

10 References

[1] Single, J.I., Schmidt, J., Denecke, J. (2020). Ontology-based computer aid for the automation of HAZOP studies. *Journal of Loss Prevention in the Process Industries*, 68, 104321. https://doi.org/10.1016/j.jlp.2020.104321.

[2] CCPS (2001). *Layer of Protection Analysis: Simplified Process Risk Assessment*. Wiley-AIChE.

[3] IEC 61882:2016. *Hazard and operability studies (HAZOP studies) – Application guide*.

A Safeguard concepts in reconstructed scenarios

The table lists each safeguard concept cited by the 40 reconstructed scenarios, the scenarios that cite it (T301-number), and the deviations concerned. These are ontology concepts reported in the source, not confirmed installed equipment.

Safeguard concept	N	Scenarios	Deviations
Collecting basin	28	001–006, 015–034, 039–040	High level; Other than composition; Elsewhere flow / loss of containment; High pressure; Low pressure; Low temperature
Define Ex protection zone	13	002, 004, 006, 016, 019, 022, 024, 026, 028, 030, 032, 034, 040	High level; Other than composition; Elsewhere flow / loss of containment; High pressure; Low pressure; Low temperature
Pressure vacuum relief valve	10	025–034	High pressure; Low pressure
Pressure relief valve	8	023–028, 035, 037	High pressure; High temperature
Level controller low alarm	6	007–012	Low level
Overfill protection	6	001–006	High level
Vacuum proof design	6	029–034	Low pressure
Dry run protection	3	007–008, 011	Low level
Periodical sample taking	3	013–015	Other than composition
Explosion proof motor	2	019, 022	Elsewhere flow / loss of containment
Ignition source assessment	2	036, 038	High temperature
Water detection system	2	015–016	Other than composition

B Abbreviations

CCPS	Center for Chemical Process Safety
Ex	Explosive (atmosphere)
HAZOP	Hazard and operability (study)
LIC	Level indicator controller
LT	Level transmitter
P&ID	Piping and instrumentation diagram
PFD	Process flow diagram